

Obligatory Exercises 1

Written by: Justas Simutis, Date: October 4, 2024

Exercise O1.1

a) $F_1 : ((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$

$$\begin{array}{c}
 \frac{\frac{q \rightarrow r, p \Rightarrow r, p}{p \rightarrow q, q \rightarrow r, p \Rightarrow r} \text{ax} \quad \frac{\frac{q, p \Rightarrow r, q}{q, q \rightarrow r, p \Rightarrow r} \text{ax} \quad \frac{q, r, p \Rightarrow r}{\Rightarrow ((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)} \text{ax}}{\frac{(p \rightarrow q) \wedge (q \rightarrow r), p \Rightarrow r}{(p \rightarrow q) \wedge (q \rightarrow r) \Rightarrow p \rightarrow r} \wedge -l} \rightarrow -l \\
 \frac{\frac{(p \rightarrow q) \wedge (q \rightarrow r) \Rightarrow p \rightarrow r}{\Rightarrow ((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)} \rightarrow -r}{\Rightarrow ((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)} \rightarrow -r
 \end{array}$$

$F_2 : \forall x \exists y (p(x) \wedge (p(y) \rightarrow q(x))) \rightarrow \forall z q(z)$

$$\begin{array}{c}
 \frac{\frac{\forall x \dots, p(a) \Rightarrow q(a), p(b)}{\forall x \dots, p(a), p(b) \rightarrow q(a) \Rightarrow q(a)} \quad \frac{\forall x \dots, p(a), q(a) \Rightarrow q(a)}{\Rightarrow \forall x \exists y (p(x) \wedge (p(y) \rightarrow q(x))) \Rightarrow \forall z q(z)} \text{ax}}{\frac{\forall x \dots, p(a), p(b) \rightarrow q(a) \Rightarrow q(a)}{\forall x \dots, \exists y p(a) \wedge (p(y) \rightarrow q(a)) \Rightarrow q(a)} \wedge -l} \rightarrow -l \\
 \frac{\frac{\forall x \dots, \exists y p(a) \wedge (p(y) \rightarrow q(a)) \Rightarrow q(a)}{\forall x \exists y (p(x) \wedge (p(y) \rightarrow q(x))) \Rightarrow q(a)} \exists -l}{\frac{\forall x \exists y (p(x) \wedge (p(y) \rightarrow q(x))) \Rightarrow \forall z q(z)}{\Rightarrow \forall x \exists y (p(x) \wedge (p(y) \rightarrow q(x))) \rightarrow \forall z q(z)} \forall -l} \rightarrow -r \\
 \frac{\frac{\forall x \exists y (p(x) \wedge (p(y) \rightarrow q(x))) \Rightarrow \forall z q(z)}{\Rightarrow \forall x \exists y (p(x) \wedge (p(y) \rightarrow q(x))) \rightarrow \forall z q(z)} \forall -r}{\Rightarrow \forall x \exists y (p(x) \wedge (p(y) \rightarrow q(x))) \rightarrow \forall z q(z)} \rightarrow -r
 \end{array}$$

Unfortunately the formula appears to be invalid since the left branch $\forall x \dots, p(a) \Rightarrow q(a), p(b)$ is a dead end which proves the formula is falsifiable. According to modus ponens if A and $A \rightarrow B$ are both true, then B is also true. This situation looks similar, but it does not appear to be the same since $p(a), p(b) \rightarrow q(a)$ is equivalent to $A, B \rightarrow C$ and not $A, A \rightarrow B$. The problem is $p(b)$ which is where I might have made a mistake.

b) $F_3 : ((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow q)$

First we attempt to prove F_3 using Sequent Calculus LK.

$$\frac{\frac{\frac{q \rightarrow r, p \Rightarrow q, p}{p \rightarrow r, q \rightarrow r, p \Rightarrow q} \text{ax} \quad \frac{\frac{r, p \Rightarrow q, q}{r, q \rightarrow r, p \Rightarrow q} \quad \frac{r, r, p \Rightarrow q}{r, q \rightarrow r, p \Rightarrow q} \rightarrow -l}{\frac{p \rightarrow r, q \rightarrow r, p \Rightarrow q}{(p \rightarrow r) \wedge (q \rightarrow r), p \Rightarrow q} \wedge -l} \rightarrow -l$$

$$\frac{\frac{(p \rightarrow r) \wedge (q \rightarrow r), p \Rightarrow q}{(p \rightarrow r) \wedge (q \rightarrow r) \Rightarrow p \rightarrow q} \rightarrow -r}{\Rightarrow ((p \rightarrow r) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow q)} \rightarrow -r$$

After failing we see exactly where the problem arises. Thus, from any of the failures we can create a counter model. We start by assigning the values to p, q, r : $I(p) = T, I(q) = F, I(r) = T$, after which we evaluate the formula.

$$\begin{aligned} v_I(p \rightarrow r) &= T \rightarrow T = T \\ v_I(q \rightarrow r) &= F \rightarrow T = T \\ v_I((p \rightarrow r) \wedge (q \rightarrow r)) &= T \wedge T = T \\ v_I(p \rightarrow q) &= T \rightarrow F = F \\ v_I(((p \rightarrow r) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow q)) &= T \rightarrow F = F \end{aligned}$$

For a formula to be valid it always has to evaluate to true. However, as shown above, under this counter model the formula evaluates to false. Therefore, the conclusion is that the formula is invalid.

Exercise O1.2

a) These are the rules for $\uparrow -left$ and $\uparrow -right$:

$$\frac{\Gamma \Rightarrow A, \Delta \quad \Gamma \Rightarrow B, \Delta}{\Gamma, A \uparrow B \Rightarrow \Delta} \uparrow -l$$

$$\frac{\Gamma, A, B \Rightarrow \Delta}{\Gamma \Rightarrow A \uparrow B, \Delta} \uparrow -r$$

These rules can be deduced from the negation and conjunction rules. That is because as defined before $A \uparrow B \equiv \neg(A \wedge B)$. Thus, by applying a negation rule first, and a conjunction rule afterwards we can arrive to the solution above.

b) The central part is the truth preservation across the interpretations. For example, if both premises of a rule are true, the conclusion must also be true. In other words, each rule must maintain semantic relationship between premises and conclusions.

To show that these two rules maintain this relationship we must show that $\Gamma \models A, \Delta$ and $\Gamma \models B, \Delta$ for $\uparrow -l$ as well as $\Gamma, A, B \models \Delta$ for $\uparrow -r$. First we assume all of these formulas to be true in some model M where both the premises and conclusions are true. Let M be any model where all formulas in Γ are true. Thus in model M both A, B and Δ are true, and since A and B are true, in $\Gamma, A, B \models \Delta$ and we assume Γ to be true, Δ is also true.